

Model-Based Software Design, A.Y. 2024/25

Assignment #1

Components of the working group (max 2 people)

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Item definition (Example)

One pedal controller

Purpose of this document

The purpose of this document is to be the input for the “Hazard Analysis and Risk Assessment” (HARA) needed to comply with the ISO26262 standard. To ensure safety, all activities of the safety life cycle have to be planned to avoid systematic failures.

Therefore, this document describes the assumption on the **one pedal acceleration/braking system** item you should develop.

An additional purpose of this document is to define and describe the item, its functionality, dependencies on, and interaction with, the driver, the environmental conditions, external measure, the boundary of the item and interfaces to other items as well as assumptions concerning other elements at the vehicle level. This document will handle the requirements and recommendations for establishing the definition of the item, including its functionality, interfaces, environmental conditions, legal requirements, and known hazards.

Purpose of the item

Please describe in this chapter the purpose of the item. Consider laws, standards, and regulations to sufficiently describe the item's purpose.

The item at our disposal aims to allow the user to take control over a vehicle by providing negative or positive torques proportional to the applied force to the throttle pedal, thus accelerating or decelerating (with the help of the regenerative braking) with no need for traditional breaking pedal, thus enabling smoother driving and enhancing the driving experience with simplified control mechanisms.

The purpose of the item is the following:

- To allow the driver to set the torque (positive→acceleration, negative→braking) applied on the driving wheels of a car. This system enables the driver to use only the throttle pedal for both the functions of accelerating or braking (up to a certain level) the vehicle. This system only allows use of the regenerative braking function of an electric/hybrid vehicle.
- As an assumption, the braking pedal is still inside the car, it acts directly on the hydraulic braking system, and its circuitry is independent of interferences from the “one” throttle/braking pedal. The information on whether the brake pedal is pressed is available for the considered item.

Functional behavior

The automatic transmission selector is implemented as a by-wire (hence, no mechanical links between the transmission and the selector are present) and features, in the order, these positions: P (park), R (reverse), N (neutral), D (drive), and B (braking/one pedal). The driver can move the transmission selector at any moment, so the actually selected mode is shown on the

dashboard screen. The item switches to the position chosen by the driver as soon as all related safety conditions are met.

The system can adopt two different behaviours, one when the automatic transmission selector (an independent system) is in the D position and the other in the B.

In particular:

- In D position mode, it reads the position of the throttle pedal and requires a traction torque proportional to the pedal position, as traditional in the automotive market. When the pedal is completely released, no torque is required meaning that the vehicle has its own braking force due to interaction with the air or the terrain, the internal combustion engine, or just the transmission power consumption due to internal frictions in the case of an electric vehicle. In this mode, to increase the braking torque, it is necessary to press the brake pedal and stop the vehicle completely. When the brake pedal is released in cars equipped with automatic transmissions, the vehicle starts to move slowly.
- In B (brake) position mode, the throttle pedal travel is divided into two regions:
 - regenerative braking, from the complete release up to a certain point (for example, 1/3 of the travel angle) that we can call the *neutral point*. The readout from the pedal inside this region is interpreted as a request for a braking torque, maximum when the pedal is completely released, then proportionally decreased upon the *neutral point*. When the pedal is released, the vehicle brakes up to completely stop its motion. From then on, the car remains stopped automatically regardless of the street slope. To make the vehicle moving, it is necessary to press the throttle pedal up to the acceleration region, described in the following, or to press the brake pedal and then release it.
 - Acceleration, from the neutral point up to the end of the travel (acceleration region), where the position is interpreted as a request of a traction torque proportional to the pedal position.

The behaviour can be described mathematically as follows:

$$\begin{cases} \tau_r = -\max(\tau_a) \cdot (1 - 3p), & \text{when } 0 < p \leq \frac{1}{3} \text{ (braking region).} \quad (1) \\ \tau_r = \max(\tau_a) \cdot \frac{3}{2} \cdot \left(p - \frac{1}{3}\right), & \text{when } \frac{1}{3} < p \leq 1 \text{ (acceleration region).} \quad (2) \end{cases}$$

where:

- τ_r is the requested torque;
- p is the pedal position expressed in normalized $[0,1]$ range;
- $\max(\tau_a)$ is the maximum acceleration torque in the forward direction.

The requested torque is positive to indicate a forward acceleration action or negative to indicate a braking (or backward) acceleration action (from here, the – sign in the equation 1).

Of course, it is still possible to use the braking pedal in case of emergencies or to increase the braking torque thanks to the hydraulic brakes.

Function	Operating elements
Determine torque request	Throttle pedal
Select transmission mode (behavior)	Automatic transmission selector
Brake pedal pressed	Data from the CAN bus regarding the status of the braking pedal

Driver notifications	Tell the driver the selected mode on the dashboard (between P, R, N, D, and B) and eventual faults. This is usually the one chosen by the by-wire selector
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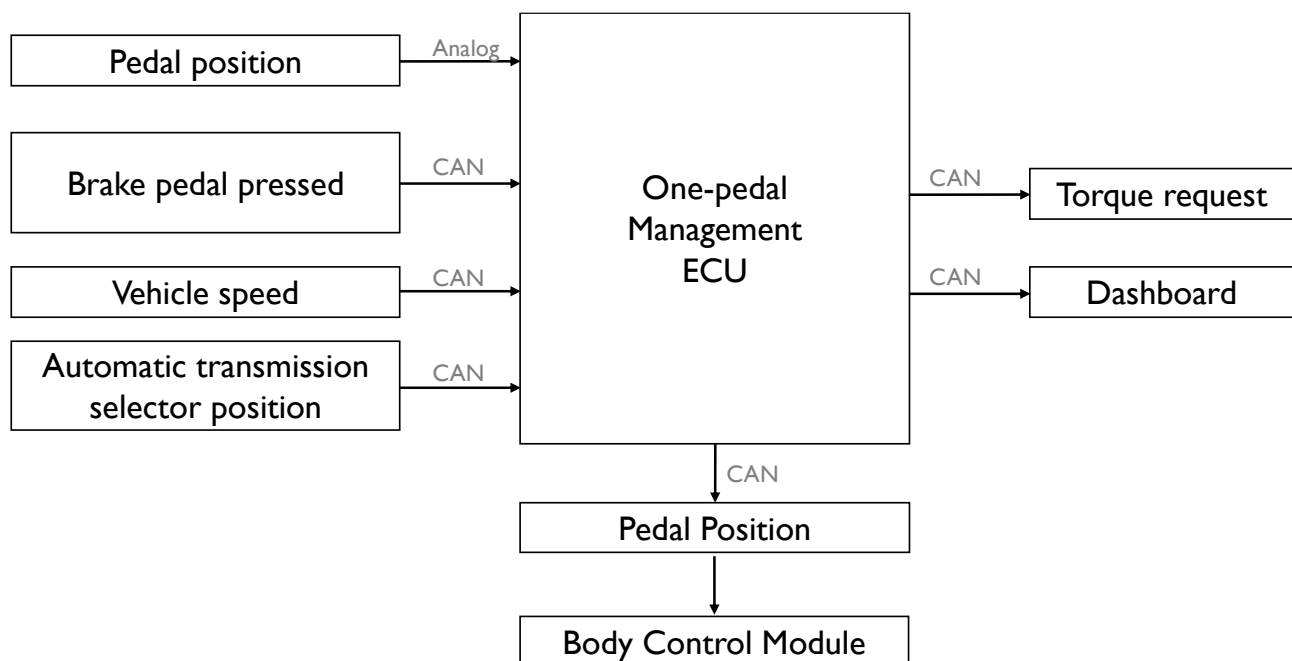
Functional block diagram

Please describe the interaction with external systems or items and/or interfaces to other elements outside the boundary of your item. Please consider the combination of “sensor-logic-actuator” and choose functional names for these elements regarding your item.

The elements of the item in hand are:

- Pedal position sensor
- A/D converter
- Microcontroller ECU
- Speed sensor
- Dashboard
- Automatic selector transmission
- CAN

In ISO 26262 we focus on the safety of electric and electronic systems, so the hazard analysis will focus on the elements listed above.



The One-Pedal Management ECU interacts with various external sensors and systems to enable and control the one-pedal driving functionality. Below is a breakdown of how it communicates with external components:

- Pedal Position Sensor: Captures the position of the pedal and sends it to the ECU for interpretation.
- Brake Pedal Pressed Signal: indicates if the brake pedal is pressed. Used for safety and logic decisions.
- Vehicle Speed Sensor: Sends current vehicle speed, which is essential for dynamic control (e.g., torque mapping, regen braking).
- Automatic transmission selector position: it provides current gear selection to the ECU to determine torque behaviour and safety logic.
- Torque Request: The ECU calculates and sends the required torque command to the powertrain system, based on inputs like pedal position and vehicle speed.
- Dashboard: Sends status and feedback information (e.g., pedal state, torque applied, system readiness) to the driver via dashboard display.
- Pedal Position Broadcast: The interpreted pedal signal is sent over the CAN network for use by other control units.
- Body Control Module: Receives processed pedal position or control signals for coordinating other vehicle body functions.

Boundaries of the system responsibility and interfaces

Please describe the boundary of the system responsibility, interaction with external systems or items and interfaces to other elements outside your item in combination with the block diagram above

The system is in charge of providing the torque request (positive or negative) to the electric motor (EM) electronic control unit (ECU).

It provides this request through the vehicular Controller Area Network (CAN).

It has to compute this torque request based on the gear selector position (negative torque only for the B position).

Moreover, it has to check the vehicle speed to determine the torque effects, in particular preventing that, during the regenerative braking, the negative torque request causes the vehicle to move in the reverse direction.

Another responsibility is to keep the vehicle stopped until the throttle pedal reaches the acceleration position and to monitor when the braking pedal is pressed to make the car slowly move when it is released.

In the reverse gear, the car acts like a standard automatic transmission car, so the vehicle only stops when the braking pedal is pressed and starts to slowly move backward when it is released.

Moreover, the transition between the position N and R or D/B is accepted only when the speed of the vehicle is lower than 5 km/h (in the same motion direction) AND the brake pedal is pressed, with the only exception on the selection of the N (neutral), which can be accepted at any time and causes the vehicle to move freewheel. The transition between R and P can be accepted only with the car almost still, and the braking pedal pressed.

Other sources of hazards, which influence the safety and reliability of the item

Please describe other sources (not E/E) of hazards, which influence the safety and reliability of the item

Loss of control due to mechanical failure of the brake pedal due wear and fatigue leading to collisions.

Over accelerated vehicles due Brake pad wear and potential accidents.

Functional requirements

Please describe all already noted functional safety requirements, this is normally output of H&R.

The system must ensure that all inputs and outputs are monitored for correctness and fail-safes are in place. The system must accurately read and validate pedal position through redundancy and plausibility checks, override torque when the brake is pressed, and monitor vehicle speed and transmission selector status to prevent unintended motion. It must also safely handle communication loss via CAN, enter safe states on critical failures, and report diagnostics to the dashboard.

Other requirements

Other environmental requirements which can influence your item

Law, directive and standard

List the laws, directives and standard which have to be considered

External measure to minimizing risks

Which external measures can be taken to minimize the risk:

- Driver's expertise in such system as it is not just an ordinary driving vehicle.
- Monitoring the behaviors of the signal pedal system to detect any faults in advance and model them to get the optimal outcome in case of any future hazards.

Hazard Analysis and Risk Assessment (Example)

One pedal controller

Participants

Name, department	Qualification	Experience
Mahdy Mokdad	BS in mechanical engineering	
Mohammad Nour edeen	BS in mechanical engineering	

Analyzes of Hazardous Situations

H1	Unintended vehicle acceleration
H2	Unintended vehicle braking
H3	Insufficient vehicle acceleration
H4	Loss of regenerative braking
H5	Unintended vehicle motion in incorrect direction
H6	Driver confusion with the dashboard display

H1; Misinformation can lead to unexpected acceleration out of the driver's control and can potentially cause a collision.

H2; Sudden deceleration can cause loss of control and can potentially cause accidents where other vehicles can collide with the rear.

H3; Certain failure in the system can interrupt the drivers' signal to accelerate (using the pedal) and lead to a delay in reaching destination.

H4; Failure to stop with minimal regenerative behavior reducing the efficiency of the system and thus beating the whole purpose of adopting this kind of system.

H5; Moving in the wrong direction, causing accidents in a confined space.

H6; Misinformation will critically endanger the user's life.

Analysis of situations

Definition of possible functional failures

Failure #	Description
F1 Pedal Position sensor failure	The sensor fails to detect the throttle pedal position.
F2 Brake pedal sensor failure	The sensor fails to detect the brake pedal press.
F3 Signal interference in CAN	Interference/Noise in the CAN causes delay in communication.
F4 Transmission selector failure	Fail to register the selected mode.
F5 Software malfunction in ECU	A bug or crash in the ECU software.
F6 Vehicle speed sensor failure	Inaccurate speed reading and improper braking/acceleration.
F7 Incorrect dashboard display	Failure in showing accurate information from the ECU.

F8 ECU overheating	Excessive load causes overheating and potential shutdown.
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Driving scenarios

Describe the possible driving situations and define the status of the vehicle you want to consider

Description of the possible driving situations

- DS1 Driving on a busy highway
- DS2 Driving on a suburban highway with minimal traffic
- DS3 Driving on a slippery road
- DS4 Driving in reverse on a busy highway
- DS5 Driving in a parking lot with tight space between cars
- DS6 Driving downhill

Definition of the vehicle status

- VS1 Driving at high speed
- VS2 Driving at low speed

Considerations

Describe driving situations for each status of the vehicle

Operational Situations #	Driving situation	Vehicle status
S1	Driving on a busy highway	VS1
S2	Driving on a busy highway	VS2
S3	Driving on a suburban highway with minimal traffic	VS1
S4	Driving on a suburban highway with minimal traffic	VS2
S5	Driving on a slippery road	VS1
S6	Driving in reverse on a busy highway	VS2
S7	Driving in a parking lot with tight space between cars	VS2
S8	Driving downhill	VS1
S9	Driving downhill	VS2

Analysis

Estimation matrix

		Risk parameters	Scenario 1	Scenario 2	Scenario 3	Scenario 4	Scenario 5	Scenario 6	Scenario 7	Scenario 8	Scenario 9	TOP EVENT
HAZARDS	H1	S	3	1	1	1	3		1	3	2	D
		E	4	3	3	3	3		3	3	3	
		C	3	1	2	1	3		2	3	1	
		ASIL	10	5	6	5	9		6	9	6	
	H2	S	3	1	2	1	2	2	1	3	2	C
		E	4	2	2	2	2	2	2	2	2	
		C	2	2	2	1	2	2	1	3	2	
		ASIL	9	5	6	4	6	6	4	8	6	
	H3	S	2	1	2	1	1		1	1	1	QM
		E	2	2	2	2	2		2	2	2	
		C	1	1	1	1	2		1	1	1	
		ASIL	5	4	5	4	5		4	4	4	
	H4	S	2	1	1	1	3		1	1	1	B
		E	2	2	2	2	2		2	2	2	
		C	2	1	1	1	3		1	2	1	
		ASIL	6	4	4	4	8		4	5	4	
	H5	S	3	1	2	1	3		1	1	1	QM
		E	1	1	1	1	1		1	1	1	
		C	2	2	2	1	2		1	1	1	
		ASIL	6	4	5	3	6		3	3	3	
	H6	S	3	1	2	1	1		1	2	1	QM
		E	1	1	1	1	1		1	1	1	
		C	2	1	2	1	2		1	2	1	
		ASIL	6	3	5	3	4		3	5	3	

Operational Situations – Comment of entries

Start with the description of what happens and then assign the parameters.

Please analyze in this way two other scenario/failure associations at your choice.

(S1) Driving on a highway with busy traffic (DS1) at a high speed (VS1) and an unintended vehicle acceleration (H1)

Effect	Loss of control that might lead to fatal collisions.	
Statement S	As the vehicle accelerates over the intended value high torques can lead to high speeds with a high relative speed with respect to other cars	S3
Statement E	Very high exposure or such a scenario as misinterpretations can rise from sensors and pedal positions.	E4
Statement C	Considering the situation at a high speed with over accelerating cars, the driver cannot control the car at such busy traffic	C3

(S8) driving downhill with a high speed (VS1) and an unintended vehicle acceleration(H1) occurs

Effect	At such a steep slope the car can easily overturn on the side of the road in a life-threatening situation.	
Statement S	High torques and high gravitational pull will lead to a tremendous increase in speed.	S3

<i>Statement E</i>	Very high exposure for such a scenario as misinterpretations can rise from sensors and pedal positions.	<i>E3</i>
<i>Statement C</i>	In case of such high speed the mechanical break will over-heat by the high friction, so the situation will be uncontrollable.	<i>C3</i>

(S1) Driving at a highway at a high speed (VS1) and an unintended vehicle break occurs(H2)

<i>Effect</i>	Sudden intense decelerations in vehicle motion can cause deadly collisions from the rear of the car.	
<i>Statement S</i>	<i>Similarly, the sudden stoppage in the car motion will certainly lead to high-speed difference.</i>	<i>S3</i>
<i>Statement E</i>	<i>Such a scenario is not likely (1% probability).</i>	<i>E2</i>
<i>Statement C</i>	<i>A brake is usually uncontrollable as it will decelerate the car almost instantaneously.</i>	<i>C3</i>

(S5) Driving on a slippery road at a high speed and an unintended vehicle acceleration occurs.

<i>Effect</i>	Inability to steer and brake effectively leads to loss of driving stability and potential accidents to occur.	
<i>Statement S</i>	At such high speed the car will be unmanageable.	<i>S3</i>
<i>Statement E</i>	1 to 10% for such a scenario as misinterpretations can rise from sensors and pedal positions.	<i>E3</i>
<i>Statement C</i>	Typically controlling a vehicle on a slippery road even with intact braking systems can be challenging.	<i>C3</i>

(S5) Driving on a slippery road at a high speed with loss of regenerative braking.

<i>Effect</i>	The delayed response of the regenerative braking system might not be sufficient.	
<i>Statement S</i>	Loss of regenerative braking weakens the overall braking system considering slippery road and the ineffectiveness of mechanical break.	<i>S3</i>
<i>Statement E</i>	Loss of regenerative brake is not a probable fault (1%)	<i>E2</i>
<i>Statement C</i>	Slippery Road.	<i>C3</i>

(S8) Driving downhill at a high speed with an unintended brake.

<i>Effect</i>	Sudden brake will cause potential harm as cars from behind can collide	
<i>Statement S</i>	Such an almost instant deceleration will lead to velocity difference and higher kinetic energy to be dissipated.	S3
<i>Statement E</i>	Low exposure (1%).	E2
<i>Statement C</i>	In such a situation no control can be taken over as the braking force is higher than the positive torque that would be provided.	C3

Safety goals

SG1	The sensor shall read accurate positions of the pedal.
SG2	The dashboard shall display the correct information.
SG3	In case of a fault in the B mode, the item shall transit to D mode alarming the driver
SG4	Loss of regenerative braking, in case it occurs, should be reported on dashboard.

Results

Failure/ Malfunction/ Hazardous situation	Safety goal	ASIL-level	Safe state	Fault tolerance time interval (FTTI)
F1&F2/unintended driving situations	SG1/SG3	D	Redundant sensors should convey correct pedal position at the same time display tat error, if occurring, on the dashboard.	100 ms
Loss of regenerative braking	SG4	B	The dashboard reports the fault and vehicle switches to D mode	70 ms

Relevant failure modes for H1

Incorrect information provided to the ECU by input signals in analog and CAN.

Relevant failure modes for H2

Glitch in the system may cause misinterpretations, forcing the ECU to take the action to suddenly turn on braking mode

Relevant failure modes for H4

In case of fully charged batteries regenerative braking can be limited by the place where the energy shall be disposed and thus reducing its functionality.

Functional Safety

Periodic check on the functionality of the microprocessor to check its internal behavior.

Use of a redundant sensor alongside the pedal position sensor to check for an error and transform to a safe state and blinking on the dashboard.